

A. Course Handout (Version 1.0)

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Algorithm Design & Implementation	Session	2025-2026
Course Code	23CS025	Semester/Batch	5 th /2023
L-T-P (Per Week)	3-0-2	Course Credits	04
Pre-requisite	Fundamentals of C Programming (23CS003), Data Structures (23CS004)	NHEQF Level	5.5
Course Coordinator	Dr. Astha Gupta	SDG Number	4,8,9

CLO01	Understand and apply the concept of algorithm complexity
CLO02	Proficiency in implementing hash tables, heaps, priority queues
CLO03	Implement the advanced sorting algorithms and apply appropriate algorithm for a particular problem
CLO04	Apply the concepts and theories of different algorithmic strategies

1. Objectives of the Course

The scope of the course is to provide the foundation for understanding the key aspects of java programming and implementation obtaining a theoretical understanding of advance programming concepts. The objectives of the course are:

- to build an understanding of analysing and evaluating algorithm efficiency.
- to inculcate the skill in students to implement and optimize data structures.
- to develop and apply advanced sorting and searching algorithms.
- to solve complex problems using greedy algorithms, backtracking, and dynamic programming.

2. Course Learning Outcomes

After completion of the course, student should be able to:

	Course Learning Outcome	*Pos	**CL	***KC	Sessions
CLO01	Understand and apply the concept of algorithm complexity	PO2, PO3, PO5, PO12	K1	Conceptual	18
CLO02	Proficiency in implementing hash tables, heaps, priority queues	PO1, PO4, PO5, PO12	K3	Conceptual Procedural	24
CLO03	Implement the advanced sorting algorithms and apply appropriate algorithm for a particular problem	PO1, PO2, PO3, PO4, PO5, PO7, PO11	K4	Conceptual Procedural	24
CLO04	Apply the concepts and theories of different algorithmic strategies	PO3, PO4, PO5, PO6, PO8	K4	Procedural	30
Total Contact Hours					96

Revised Bloom's Taxonomy Terminology

* PO's available at (shorturl.at/cryzF)

**Cognitive Level =CL

***Knowledge Categories = KC

Course Learning Outcomes	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO01		H	H		H	M						H
CLO02	H			H	H	M		M				H
CLO03	H	H	H	H	H		H				H	
CLO04			H	H	H		H				H	

H=High, M=Medium, L=Low

3. ERISE Grid Mapping

Feature Enablement	Level (1-5, 5 being highest)
Entrepreneurship	1
Research	2
Innovation	4
Skills	5
Employability	5

4. Recommended Books (Reference Books/Textbooks)

B01: "Classic Data Structures", Samanta and Debasis, 2nd edition, PHI publishers.

B02: "Fundamentals of Data Structures", Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.

B03: "Data Structures with C (Schaum's Outline Series)", Seymour Lipschutz, 1st edition, McGraw Hill Education.

B04: "Design and Analysis of Algorithms" by S. Sridhar, Oxford University Press, 2014.

B05: "Algorithms, Data Structures, and Problem Solving with C++", Illustrated Edition by Mark Allen Weiss, Addison-Wesley Publishing Company.

B06: "Design and Analysis of Algorithms" by S. Sridhar, Oxford University Press, 2014.

5. Other readings & relevant websites

S.N.	Link of Journals, Magazines, Websites, and Research Papers
1.	https://www.freecodecamp.org/news/big-o-cheat-sheet-time-complexity-chart
2.	https://www.geeksforgeeks.org/distributing-m-items-circle-size-n-starting-k-th-position/
3.	https://www.geeksforgeeks.org/open-addressing-collision-handling-technique-in-hashing/
4.	https://www.geeksforgeeks.org/find-the-winner-of-the-game-2/
5.	https://www.geeksforgeeks.org/priority-queue-using-linked-list/
6.	https://www.youtube.com/watch?v=HqPJF2L5h9U

6. Recommended Tools and Platforms

Code Quotient

7. Course Plan

Lecture Number	Topics	Recommended Books
1-8	Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II. Practice Problem: Prime Factorization, GCD of two numbers, Distribute in circle	BO1, BO2
9-18	Problem Solving Techniques: Brute Force, Greedy Algorithms, Divide and Conquer, Dynamic Programming, Backtracking, Randomized Algorithms, Recursion, Two Pointer Technique, Sliding Window Technique	BO2
19-24	Backtracking: Permutations and Combinations, Subset Sum and Partition, Word Search, Hamiltonian Path. Practice Problem: Robot Movement, Solve Sudoku, Rat in a Maze, Print all strings of n-bit.	BO3
25-30	Tree: Introduction, BST, Types of Trees, Tree Traversal and Operations, HBT(AVL) – I, HBT(AVL) – II. Practice Problem: Find the lowest common ancestor of a given two nodes in a binary search tree, Find the kth smallest element in the binary search tree.	BO3
ST-1 (Syllabus covered from Lecture 1 to 30)		
31-34	Heap: Min Heap, Max Heap, Operations on Heap, Priority Queues Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, Check if a given tree is max-heap or not	
35-48	Greedy Algorithms: Introduction, Fractional Knapsack, Job Scheduling, Bellman Ford.	BO2
49-60	Dynamic Programming: Introduction, Memorization, Optimal Substructure and Overlapping Subproblem, 0/1 Knapsack, Matrix Chain Multiplication, Travelling Salesman Problem, Longest Common Subsequence. Practice Problem: Longest Common Subsequence (LCS), Count number of ways to cover a distance, Matrix Chain Multiplication problem, 0-1 Knapsack problem.	BO4
61-74	Graph: Introduction, Types, Traversal - BFS and DFS, Prim's Algorithm, Kruskal's Algorithm. Practice Problem: Find the cycle in an undirected graph, Find the minimum number of edges in a path of a graph, find path in a directed graph, Find the number of islands.	BO5
ST-2 (Syllabus covered from Lecture 31 to 74)		
75-82	Greedy Algorithms (Advanced): Dijkstra, Bellman-Ford Practice Problems: Interval Scheduling, Job Scheduling with Deadlines	BO2
83-96	Hashing: Intro, Collision Handling Techniques Practice Problem: Noise In the Library, try balancing the scale, Find out the winner	BO6
END TERM – FULL SYLLABUS		

8. Delivery/Instructional Resources

Lecture No.	Topics	Web References	Audio-Video
1-8	Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II.	https://www.freecodecamp.org/news/big-o-cheat-sheet-time-complexity-chart	https://www.youtube.com/watch?v=HfIH3czXc-8
9-18	Brief Introduction about various Problem-Solving techniques such as, Brute Force, Greedy Algorithms, Divide and Conquer, Dynamic Programming, Backtracking, Randomized Algorithms, Recursion, Two Pointer Technique, Sliding Window Technique	https://www.geeksforgeeks.org/	https://rb.gy/mm1kzl
19-24	Backtracking: Permutations and Combinations, Subset Sum and Partition, Word Search, Hamiltonian Path. Practice Problem: Robot Movement, Solve Sudoku, Rat in a Maze, Print all strings of n-bit.	https://www.prepbytes.com/blog/backtracking/	https://www.youtube.com/watch?v=xFv_Hl4B83A
25-30	Tree: Introduction, BST, Types of Trees, Tree Traversal and Operations, HBT(AVL) – I, HBT(AVL) – II. Practice Problem: Find the lowest common ancestor of a given two nodes in a binary search tree, Find the kth smallest element in the binary search tree.	https://www.scaler.com/topics/traversal-of-binary-tree/	https://www.youtube.com/watch?v=I_JuQ5ayPmc
31-34	Heap: Min Heap, Max Heap, Operations on Heap, Priority Queues Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, Check if a given tree is max-heap or not	https://www.geeksforgeeks.org/priority-queue-using-linked-list/	https://www.youtube.com/watch?v=HqPJF2L5h9U
35-48	Greedy Algorithms: Introduction, Fractional Knapsack, Job Scheduling, Bellman Ford.	https://www.geeksforgeeks.org/fractional-knapsack-problem/	https://www.youtube.com/watch?v=oTTzN MHM05I
49-60	Dynamic Programming: Introduction, Memorization, Optimal Substructure and Overlapping Subproblem, 0/1 Knapsack, Matrix Chain Multiplication, Travelling Salesman Problem, Longest Common Subsequence. Practice Problem: Longest Common Subsequence (LCS), Count number of ways to cover a distance, Matrix Chain Multiplication problem, 0-1 Knapsack problem.	https://www.geeksforgeeks.org/tabulation-vs-memoization/	https://rb.gy/pjrlrv

61-74	Graph: Introduction, Types, Traversal - BFS and DFS, Prim's Algorithm, Kruskal's Algorithm. Practice Problem: Find the cycle in an undirected graph, Find the minimum number of edges in a path of a graph, find path in a directed graph, Find the number of islands.	https://www.geeksforgeeks.org/difference-between-bfs-and-dfs/	https://www.youtube.com/watch?v=N2P7w22tN9c
75-82	Greedy Algorithms (Advanced): Dijkstra, Bellman-Ford Practice Problems: Interval Scheduling, Job Scheduling with Deadlines	https://www.geeksforgeeks.org/dsa/dijkstras-shortest-path-algorithm-greedy-algo-7/	https://www.youtube.com/watch?v=bZkzH5x0SKU
83-96	Hashing: Intro, Collision Handling Techniques Practice Problem: Noise In the Library, try balancing the scale, Find out the winner	https://www.geeksforgeeks.org/open-addressing-collision-handling-technique-in-hashing/	https://www.youtube.com/watch?v=W5q0xgxmRd8&list=PLxM5rx4f4fwOPORqEZZhAA Y5OG0WMZff

9. Lab Plan

Lab Number	Practical	Learning Resources
1-8	Introduction to Time and Space Complexity: 1. Find the prime factors of a given number. 2. Compute GCD of two numbers using Euclidean Algorithm. 3. Distribute N items in a circle and find who gets the last one	https://www.youtube.com/watch?v=5gFC-ayyQMk
9-18	Miscellaneous Problem: 1. Solve a problem using brute force. 2. Apply the greedy approach to interval scheduling. 3. Use divide and conquer to sort an array. 4. Solve 0/1 Knapsack using dynamic programming. 5. Generate permutations of a string using backtracking. 6. Give one example of a randomized algorithm. 7. Solve Tower of Hanoi using recursion. 8. Find a pair with sum K using two pointers. 9. Find the max sum subarray of size k using sliding window.	https://rb.gy/mm1kzl https://rb.gy/s5tk7c
19-24	Backtracking: 1. Move a robot into a grid using backtracking. 2. Solve Sudoku using backtracking. 3. Find all paths for Rat in a Maze. 4. Print all binary strings of n bits.	https://rb.gy/hdua59
25-30	Tree: 1. Find the lowest common ancestor of a given two nodes in a binary search tree 2. Find the kth smallest element in the binary search tree	https://rb.gy/mm1kzl

31-34	Heap: 1. Find max/min in the continuous stream of data 2. Sort an array using heap sort 3. Check if a given tree is max-heap or not	https://www.youtube.com/watch?v=NKJnHewiGdc
35-48	Greedy Algorithms: 1. Solve the fractional knapsack problem. 2. Solve a graph with negative weights using Bellman-Ford.	https://www.youtube.com/watch?v=oTTzNMHM05I
49-60	Dynamic Programming: 1. Longest Common Subsequence (LCS) 2. Count ways to cover distance N with 1, 2, or 3 steps. 3. Solve Matrix Chain Multiplication using DP. 4. Solve the 0/1 Knapsack problem using DP.	https://www.youtube.com/watch?v=PfkBS9qIMRE
61-74	Graph: 1. Find the cycle in an undirected graph 2. Find the minimum number of edges in a path of a graph 3. Find path in a directed graph 4. Find the number of islands	https://www.youtube.com/watch?v=porShXfpPqA
75-82	Greedy Algorithms (Advanced): 1. Find shortest path using Dijkstra's algorithm. 2. Solve the interval scheduling problem. 3. Schedule jobs with deadlines to maximize profit.	https://www.youtube.com/watch?v=zPtI8q9gvX8
83-96	Hashing: 1. Solve the "Noise in the Library" problem using hashing. 2. Use hashing to balance a scale with given weights. 3. Find the winner in a voting system using hashing.	https://www.youtube.com/watch?v=8uFWG6xfkuc

10. Action plan for different types of learners

Slow Learners	Average Learners	Fast Learners
● Remedial Classes ● Encouragement for improvement using Peer Tutoring ● Use of Audio and Visual Materials ● Use of Real-Life Examples	● Expert Lecture ● Formative Exercises used to highlight concepts and notions ● E-notes and E-exercises to read ahead of the pedagogic material.	● Engaging students to hold hands of slow learners by creating a Peer Tutoring Group ● Design solutions for complex problems ● Design solutions for complex problems ● Presentation on topics beyond those covered in CHO

11. Evaluation Scheme & Components

Evaluation Component	Type of Component	No. of Assessments	Weightage of Component	Mode of Assessment
----------------------	-------------------	--------------------	------------------------	--------------------

Component 2	Sessional Tests (STs)	02*	40%	Online/On Campus
Component 3	End Term Examination	01	60%	Online/On Campus
Total		100%		

* Out of 02 STs, best 1 ST for final marks evaluation of STs will be considered.

* Makeup Examination will compensate for either ST-1 or ST-2 (Only for genuine cases, based on the Dean's approval and the exam will be based on the full syllabus).

**As per Academic Guidelines, a minimum of 85% attendance is required to become eligible for appearing in the End Semester Examination.

12. Syllabus of the Course

Subject: Algorithm Design & Implementation	Subject Code: 23CS025	
Contents	Lectures	Weightage (%)
Introduction to the Course: Overview of advance data structure, Introduction to Time and Space Complexity: Iterative and Recursive Approach – I, Iterative and Recursive Approach – II.	8	6%
Tree: Introduction, BST, Types of Trees, Tree Traversal and Operations, HBT(AVL) – I, HBT(AVL) – II.	10	10%
Heap: Min Heap, Max Heap, Operations on Heap, Priority Queues. Practice Problem: Find max/min in the continuous stream of data, sort an array using heap sort, Check if a given tree is max-heap or not.	6	8%
Graph: Introduction, Types, Traversal - BFS and DFS, Prim's Algorithm, Kruskal's Algorithm.	6	10%
Hashing: Intro, Collision Handling Techniques.	4	10%
Brief Introduction about various Problem-Solving techniques such as, Brute Force, Greedy Algorithms, Divide and Conquer, Dynamic Programming, Backtracking, Randomized Algorithms, Recursion, Two Pointer Technique, Sliding Window Technique	14	16%
Greedy Algorithms: Introduction, Fractional Knapsack, Job Scheduling, Dijkstra, Bellman Ford.	12	10%
Backtracking: Permutations and Combinations, Subset Sum and Partition, Word Search, Hamiltonian Path.	14	12%
Dynamic Programming: Introduction, Memorization, Optimal Substructure and Overlapping Subproblem, 0/1 Knapsack, Matrix Chain Multiplication, Travelling Salesman Problem, Longest Common Subsequence.	22	18%
End Term 100% Syllabus		

This document is approved by

Designation	Name	Signature
Course Coordinator	Dr. Astha Gupta	
Head Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date (DD/MM/YYYY)	14/10/2025	